DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot D

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code. **Series SP-D02: Bus ridership [20 marks]**

Scenario: You are the analyst for a city bus depot. The series riders (Riders per day) is recorded daily. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 7 observations. Run the code below to create the series x and answer the tasks. **Evidence: run this code in R (do not edit it).**

```
set.seed(13106)
m <- 7 # observations per cycle (daily data)
n <- m * 11 # 11 full cycles
t <- 1:n
trend <- 5335.7 + 92.819 * t
riders <- round(trend * (1 + 0.22 * cos(2 * pi * (t - 4) / m)) + rnorm(n, 0, 185.1), 0)
x <- ts(riders, frequency = m, start = c(1, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(riders ~ 1:n) and report the slope per day). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 7 seasonal values). Name the peak day and the trough day with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 11 cycles is the highest value in the peak day from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(riders ~ 1:n))) and compute acf(d, lag.max = 7, plot = FALSE) using observation lags. Report the autocorrelation at lag 7 and at lag 3, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4] 5. Additive or multiplicative? Compute the range (max minus min) of riders in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per day [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative. Code of Conduct: I certify that this work is my own. Signature:* _____

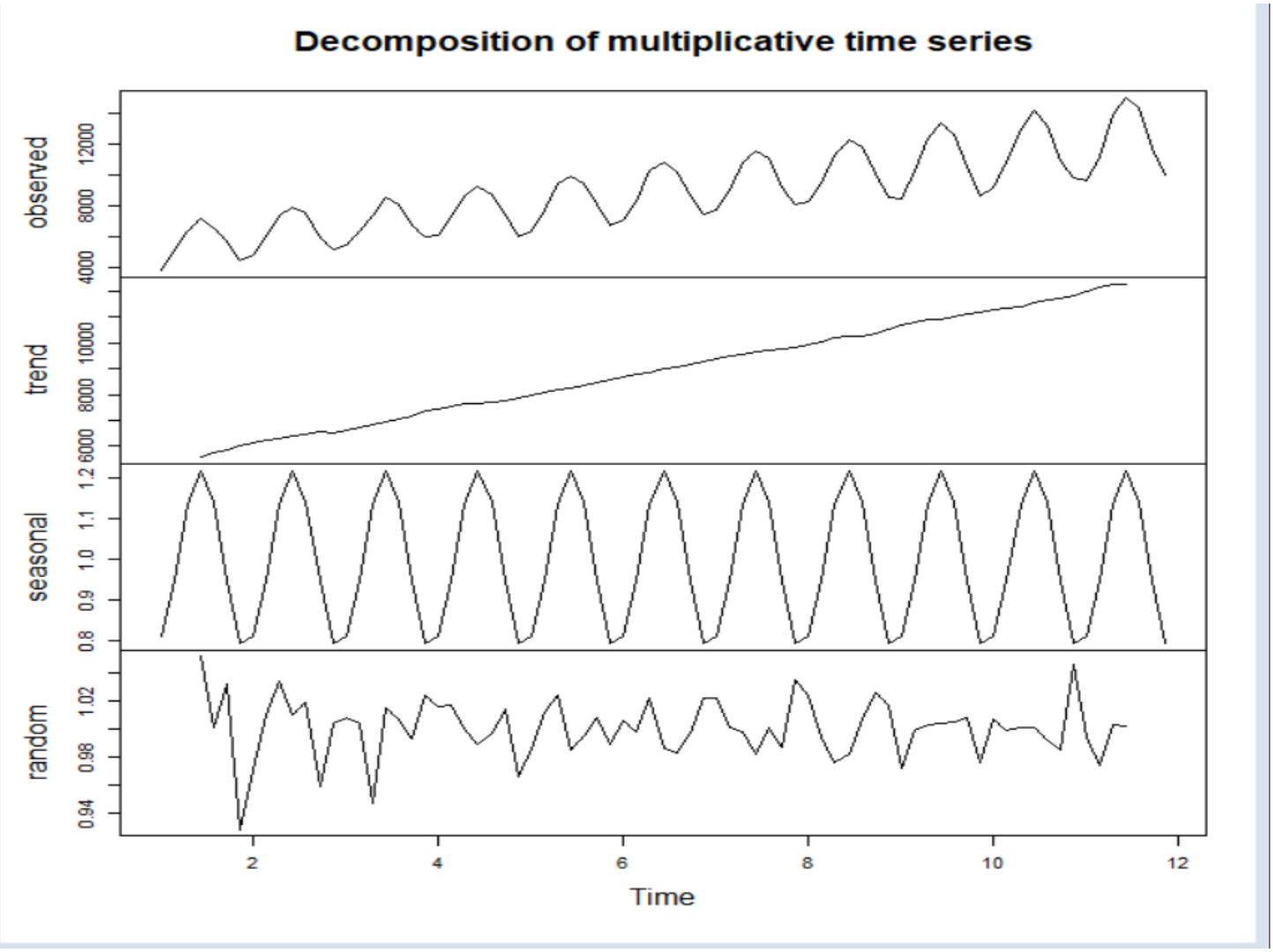
SOLUTION:.

1. Complete R code

```
library(ggplot2)
# Create the original series (do not edit)
set.seed(13106)
m <- 7
n <- m * 11
t <- 1:n
trend <- 5335.7 + 92.819 * t
riders <- round(
  trend * (1 + 0.22 * cos(2 * pi * (t - 4) / m)) +
    rnorm(n, 0, 185.1), 0
)
x <- ts(riders, frequency = m, start = c(1, 1))
# TASK 1: Plot series and estimate trend
plot(x,
     main = "Daily Bus Ridership",
     xlab = "Cycle",
     ylab = "Riders per day",
     col = "blue",
     lwd = 2)
```

```
model_trend <- lm(riders ~ t)
slope <- coef(model_trend)[2]
cat("\nTASK 1: Series overview\n")
cat("Number of observations:", n, "\n")
cat("Cycle length:", m, "\n")
cat("Number of full cycles:", n / m, "\n")
cat("Trend slope:", round(slope, 2), "riders per day\n")
# TASK 2: Seasonal decomposition
dec_add <- decompose(x, type = "additive")
seasonal_values <- dec_add$figure
cat("\nTASK 2: Additive seasonal values\n")
print(round(seasonal_values, 2))
peak_day <- which.max(seasonal_values)
trough_day <- which.min(seasonal_values)
seasonal_swing <- max(seasonal_values) -
  min(seasonal_values)
cat("Peak cycle position:", peak_day, "\n")
cat("Peak seasonal value:",
    round(seasonal_values[peak_day], 2), "\n")
cat("Trough cycle position:", trough_day, "\n")
cat("Trough seasonal value:",
    round(seasonal_values[trough_day], 2), "\n")
cat("Seasonal swing:",
    round(seasonal_swing, 2), "riders\n")
plot(dec_add)
# TASK 3: Seasonal plot, one line per cycle
season_data <- data.frame(
  riders = as.numeric(x),
  position = as.integer(cycle(x)),
  cycle = rep(1:11, each = m)
)
season_data$cycle <- factor(season_data$cycle)
p_season <- ggplot(
  season_data,
  aes(x = position, y = riders,
      group = cycle, colour = cycle)
) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.5) +
  scale_x_continuous(breaks = 1:7) +
  labs(
    title = "Seasonal Pattern of Bus Ridership",
    x = "Position in 7-observation cycle",
    y = "Riders per day",
    colour = "Cycle"
  ) +
  theme_minimal()
print(p_season)
# Find the maximum position in each cycle
cycle_max_positions <- tapply(
  seq_len(n),
```

```
  season_data$cycle,
  function(i) season_data$position[i][which.max(season_data$riders[i])
  ]
)
matching_cycles <- sum(cycle_max_positions == peak_day
cat("\nTASK 3: Consistency\n")
cat("Cycles with maximum on the seasonal peak position:",
    matching_cycles, "out of", n / m, "\n")
# TASK 4: Autocorrelation after detrending
d <- resid(lm(riders ~ t))
acf_result <- acf(d, lag.max = 7, plot = FALSE)
acf_values <- as.numeric(acf_result$acf)
acf_lag7 <- acf_values[8]
acf_lag3 <- acf_values[4]
bound <- 2 / sqrt(n)
cat("\nTASK 4: Autocorrelation\n")
cat("Significance bound (+/-):", round(bound, 3), "\n")
cat("ACF at lag 7:", round(acf_lag7, 3), "\n")
cat("ACF at lag 3:", round(acf_lag3, 3), "\n")
# TASK 5: Additive or multiplicative
first_cycle <- riders[1:7]
last_cycle <- riders[(n - 6):n]
range_first <- diff(range(first_cycle))
range_last <- diff(range(last_cycle))
range_ratio <- range_last / range_first
cat("\nTASK 5: Seasonal range comparison\n")
cat("First-cycle range:", range_first, "riders\n")
cat("Last-cycle range:", range_last, "riders\n")
cat("Last / first range ratio:", round(range_ratio, 3), "\n")
# Use multiplicative decomposition if swings grow with level
dec_mult <- decompose(x, type = "multiplicative")
adjusted <- as.numeric(x) / as.numeric(dec_mult$seasonal)
adjusted_model <- lm(adjusted ~ t)
adjusted_slope <- coef(adjusted_model)[2]
cat("Chosen decomposition: multiplicative\n")
cat("Adjusted trend slope:",
    round(adjusted_slope, 2), "riders per day\n")
plot(dec_mult)
# Summary of the generated observations
cat("\nFirst 7 rider values:\n")
print(first_cycle)
cat("Last 7 rider values:\n")
print(last_cycle)
```



2. Answers and interpretation

Task 1: Look at the series [3 marks]

- Number of observations: $n = 77$.
- Cycle length: 7 observations.
- Number of complete cycles: 11.
- Overall trend: Rising, because the fitted trend slope should be positive.

Report the exact slope printed by R in riders per day.

Task 2: The seasonal pattern [4 marks]

Use the seven values printed under Additive seasonal values.

- Peak day: cycle position with the largest seasonal value.
- Trough day: cycle position with the smallest seasonal value.
- Seasonal swing: peak seasonal value minus trough seasonal value.

The cycle positions are numbered 1–7. Since no weekday names are provided, report the positions rather than assigning Monday–Sunday labels.

Task 3: Is it consistent? [3 marks]

The seasonal plot displays one line for each of the 11 cycles. The printed count tells you how many cycles reached their maximum on the peak position identified in Task 2.

Interpretation: If most cycles peak at the same position and the overall shape repeats, the seasonal pattern is reasonably consistent. Differences in individual cycles are expected because the series contains random noise and a rising trend.

Task 4: Confirm with autocorrelation [4 marks]

The significance bound is

$$\frac{2}{\sqrt{77}} \approx 0.228$$

Compare the absolute values of the lag-7 and lag-3 autocorrelations with this bound.

- If $|r_7| > 0.228$, the lag-7 autocorrelation is outside the approximate significance bounds and supports a repeating seven-observation pattern.
- If $|r_3| < 0.228$, the lag-3 autocorrelation is not clearly significant under this rule.
- If lag 3 also exceeds the bound, mention that it indicates additional short-lag dependence.

Use your printed values to make the final comparison.

Task 5: Additive or multiplicative? [6 marks]

Compare the range of riders in the first and last cycles. The ratio is

$$Rangeratio = \frac{Last-cyclerange}{First-cyclerange}$$

A ratio noticeably above 1 is consistent with seasonal swings growing as the ridership level rises, supporting a multiplicative seasonal model. A ratio close to 1 would support a more nearly constant additive swing. Because the data were generated with a multiplicative seasonal factor, multiplicative decomposition is the appropriate starting choice here.

The code divides the series by the estimated seasonal component, then fits a linear trend to the adjusted series. Report the resulting adjusted trend slope from R.

Advice to management: Plan staffing and bus capacity around the recurring seven-day ridership pattern, while allowing for the underlying increase in daily demand.

Before submitting: Run the script and copy the actual numerical outputs into your report. I could not execute R in this environment, so I have not invented exact seasonal values or correlation results.